

All of Statistics: A Concise Course in Statistical Inference
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All errors, typographical and substantive, and other offenses, are entirely my own.

Chapter 4: Inequalities

Import Packages

Please see the associated GitHub repo for all code and comments to simulations.
[LINK HERE]

Question: 4.5.1

Let $X \sim \text{Exp}(\beta)$. Find $\mathbb{P}(|X - \mu_X| \geq k\sigma_X)$ for $k > 1$. Compare this to the bound you get from Chebyshev's inequality.

For $X \sim \text{Exp}(\beta)$, $\mu_X = \beta$ and $\sigma_X^2 = \beta^2$. Calculating the probability exactly,

$$\begin{aligned}\mathbb{P}(|X - \mu_X| \geq k\sigma_X) &= \mathbb{P}(X - \mu_X \geq k\beta) \\ &= \mathbb{P}(X \geq (k+1)\beta) \\ &= \frac{1}{\beta} \int_{(k+1)\beta}^{\infty} \exp(-x/\beta) \, dx \\ &= -\exp(-x/\beta) \Big|_{(k+1)\beta}^{\infty} \\ &= \exp(-(k+1))\end{aligned}$$

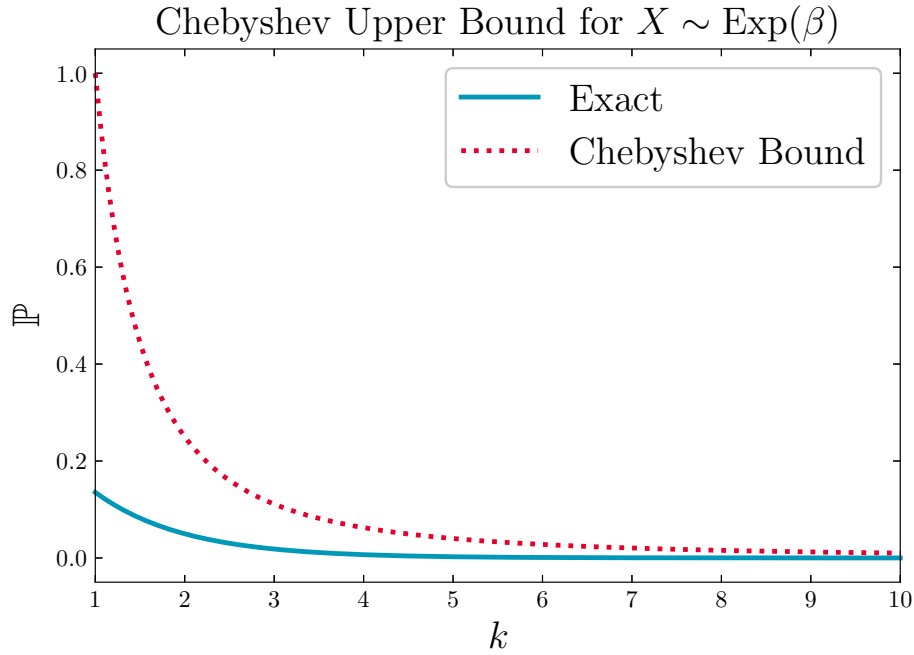
with the first equality following from the fact that since $x > 0$, $\mu = \sigma_X = \beta$, no x can be $k\sigma_X = k\beta$ to the left of the mean.

By Chebyshev's inequality,

$$\mathbb{P}(|X - \beta| \geq k\sigma_X) \leq \frac{\sigma_X^2}{k^2 \sigma_X^2} = \frac{1}{k^2}$$

where $\exp(-(k+1)) < 1/k^2$ for $k > 1$.

Here we graph the exact probability and the Chebyshev bound as functions of k :

**Question: 4.5.2**

Let $X \sim \text{Poisson}(\lambda)$. Use Chebyshev's inequality to show that $\mathbb{P}(X \geq 2\lambda) \leq 1/\lambda$.

PROOF. For $X \sim \text{Poisson}(\lambda)$, we have $\mu_X = \sigma_X^2 = \lambda$. In the case where $X - \lambda \geq 0$ and $t = \lambda$, we have

$$\begin{aligned} \mathbb{P}(|X - \lambda| \geq \lambda) &= \mathbb{P}(X - \lambda \geq \lambda) \\ &= \mathbb{P}(X \geq 2\lambda) \end{aligned}$$

Note that when $X - \lambda < 0$ with $t = \lambda$, we have

$$\begin{aligned} \mathbb{P}(|X - \lambda| \geq t) &= \mathbb{P}(-(X - \lambda) \geq \lambda) \\ &= \mathbb{P}(X \leq \lambda - \lambda) \\ &= \mathbb{P}(X \leq 0) \\ &= 0 \end{aligned}$$

so we need only consider the case where $X - \lambda \geq 0$. Applying Chebyshev's inequality, we can conclude

$$\mathbb{P}(X \geq 2\lambda) \leq \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

□

Question: 4.5.3

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ and $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Bound $\mathbb{P}(|\bar{X}_n - p| > \epsilon)$ using Chebyshev's inequality and using Hoeffding's inequality. Show that, when n is large, the bound from Hoeffding's inequality is smaller than the bound from Chebyshev's inequality.

PROOF. For $\bar{X}_n$, we have $\mathbb{E}(\bar{X}_n) = p$ and $\mathbb{V}(\bar{X}_n) = p(1-p)/n$. The upper bounds from Chebyshev and Hoeffding are

$$\mathbb{P}(|\bar{X}_n - p| > \epsilon) \leq \frac{p(1-p)}{n\epsilon^2} \quad (\text{Chebyshev})$$

$$\mathbb{P}(|\bar{X}_n - p| > \epsilon) \leq 2e^{-2n\epsilon^2} \quad (\text{Hoeffding})$$

Taking the ratio of the Hoeffding to Chebyshev limits, we aim to calculate

$$\lim_{n \rightarrow \infty} \frac{2e^{-2n\epsilon^2}}{p(1-p)/n\epsilon^2} = \frac{1}{p(1-p)} \lim_{n \rightarrow \infty} \frac{2n\epsilon^2}{e^{2n\epsilon^2}}$$

Using l'Hôpital's rule to evaluate the limit since $\lim_{n \rightarrow \infty} 2n\epsilon^2 = +\infty$ and $\lim_{n \rightarrow \infty} e^{2n\epsilon^2} = +\infty$, we have

$$\frac{1}{p(1-p)} \lim_{n \rightarrow \infty} \frac{2n\epsilon^2}{e^{2n\epsilon^2}} = \frac{1}{p(1-p)} \lim_{n \rightarrow \infty} \frac{2\epsilon^2}{2\epsilon^2 e^{2n\epsilon^2}} = \frac{1}{p(1-p)} \lim_{n \rightarrow \infty} \frac{1}{e^{2n\epsilon^2}} = 0$$

ascertaining that Hoeffding is a tighter bound than Chebyshev. $\square$

Question: 4.5.4

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

- (a) Let $\alpha > 0$ be fixed and define

$$\epsilon_n = \sqrt{\frac{1}{2n} \log \left(\frac{2}{\alpha} \right)}.$$

Let $\hat{p}_n = n^{-1} \sum_{i=1}^n X_i$. Define $C_n = (\hat{p}_n - \epsilon_n, \hat{p}_n + \epsilon_n)$. Use Hoeffding's inequality to show that

$$\mathbb{P}(C_n \text{ contains } p) \geq 1 - \alpha.$$

In practice, we truncate the interval so it does not go below 0 or above 1.

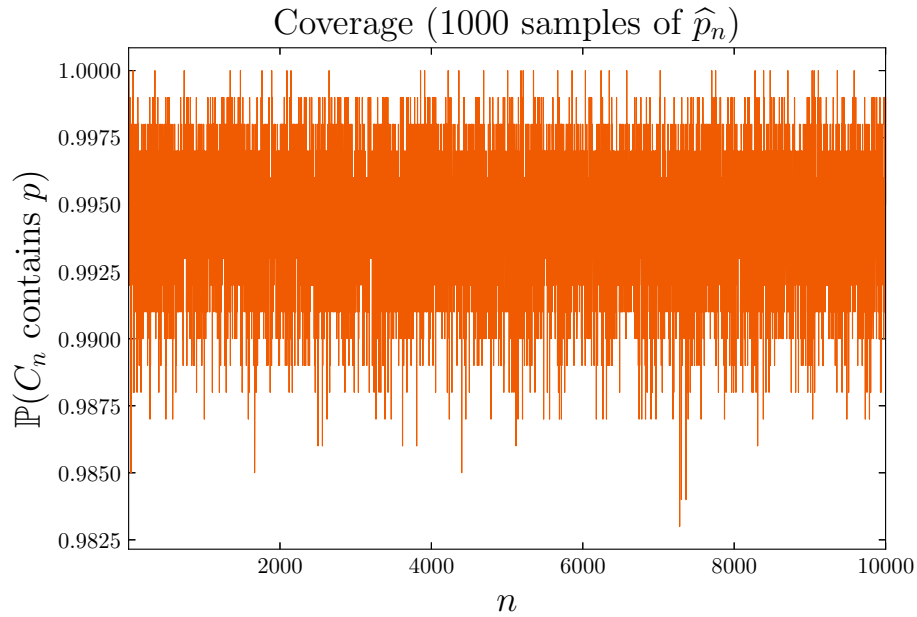
- (b) (Computer Experiment.) Let's examine the properties of this confidence interval. Let $\alpha = 0.05$ and $p = 0.4$. Conduct a simulation study to see how often the interval contains p (called the coverage). Do this for various values of n between 1 and 10000. Plot the coverage versus n .
- (c) Plot the length of the interval versus n . Suppose we want the length of the interval to be no more than .05. How large should n be?

PROOF. (a) Since $\bar{X}_n = \hat{p}_n$ and $\mathbb{P}(|\bar{X}_n - p| > \epsilon_n) \leq \alpha$, by complementary events (either $|\bar{X}_n - p| > \epsilon_n$ or $|\bar{X}_n - p| \leq \epsilon_n$), we have

$$\mathbb{P}(|\bar{X}_n - p| \leq \epsilon_n) \geq 1 - \alpha$$

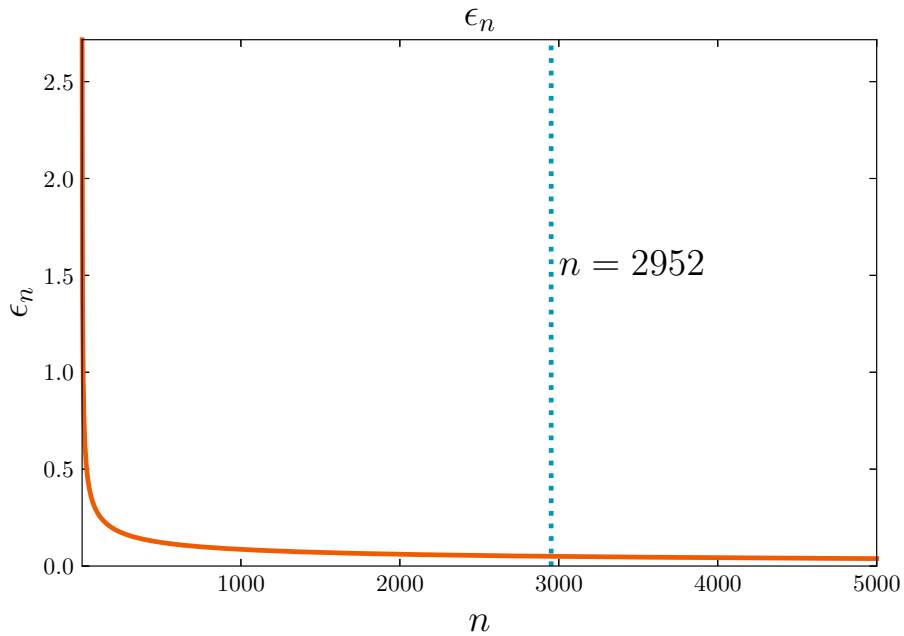
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(b) Here, we sample numerous values of $\hat{p}_n$, construct C_n from each value, and determine if $p \in C_n$.



As $n \rightarrow +\infty$, we see that the coverage asymptotically approaches 1. Intuitively this is sensible. Since $X_i \sim \text{Bernoulli}(p)$, for large n , $\hat{p}_n \rightarrow p$ (this is the weak law of large numbers, covered in the next chapter).

(c) The length of C_n is $2\epsilon_n$, and for the length to be no more than 0.05, have $n \geq 2952$.



Question: 4.5.5

Prove Mill's inequality, Theorem 4.7. Hint. Note that $\mathbb{P}(|Z| > t) = 2\mathbb{P}(Z > t)$. Now write out what $\mathbb{P}(Z > t)$ means and note that $x/t > 1$ whenever $x > t$.

Theorem (Wasserman 4.7) (Mill's Inequality)

Let $Z \sim N(0, 1)$. Then

$$\mathbb{P}(|Z| > t) \leq \sqrt{\frac{2}{\pi}} \frac{e^{-t^2/2}}{t}.$$

PROOF. By symmetry of the standard normal distribution,

$$\mathbb{P}(|Z| > t) = 2\mathbb{P}(Z > t)$$

Then

$$\begin{aligned} \mathbb{P}(|Z| > t) &= 2\mathbb{P}(Z > t) \\ &= 2 \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \end{aligned}$$

Now, multiply both sides by t :

$$\begin{aligned} \implies t\mathbb{P}(|Z| > t) &= \sqrt{\frac{2}{\pi}} t \int_t^\infty \exp\left(-\frac{z^2}{2}\right) dz \\ &\leq \sqrt{\frac{2}{\pi}} \int_t^\infty z \exp\left(-\frac{z^2}{2}\right) dz \\ &= \sqrt{\frac{2}{\pi}} e^{-t^2/2} \end{aligned}$$

which lastly implies that $\mathbb{P}(|Z| > t) \leq \sqrt{\frac{2}{\pi}} e^{-t^2/2}/t$. $\square$

Question: 4.5.6

Let $Z \sim N(0, 1)$. Find $\mathbb{P}(|Z| > t)$ and plot this as a function of t . From Markov's inequality, we have the bound $\mathbb{P}(|Z| > t) \leq \frac{\mathbb{E}|Z|^k}{t^k}$ for any $k > 0$. Plot these bounds for $k = 1, 2, 3, 4, 5$ and compare them to the true value of $\mathbb{P}(|Z| > t)$. Also, plot the bound from Mill's inequality.

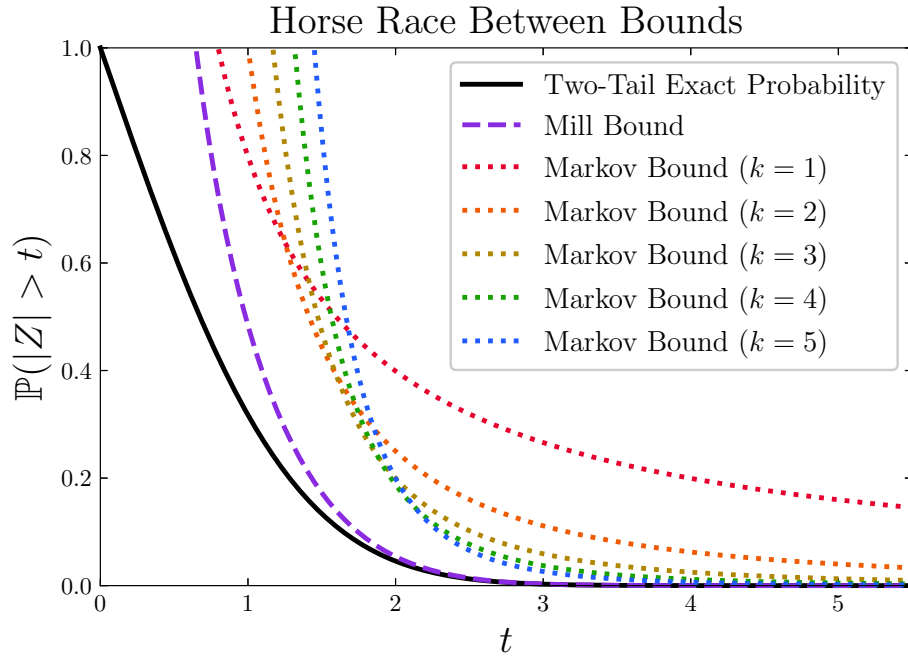
Some initial observations to bear in mind:

- Observe that $\mathbb{P}(|Z| > t)$ is the two-tailed probability that we observe $Z > t$, when $Z \geq 0$, or $Z < -t$, when $Z < 0$.
- Using Markov's inequality, $\mathbb{P}(|Z| > t) = \mathbb{P}(|Z|^k > t^k) \leq \mathbb{E}|Z|^k / t^k$.

- The expectation term $\mathbb{E}(|Z|)$ is defined as

$$\mathbb{E}|Z| = \int_0^{+\infty} z f_Z(z) dz + \int_{-\infty}^0 z f_Z(z) dz = 2 \int_0^{+\infty} z f_Z(z) dz = \sqrt{\frac{2}{\pi}}$$

Running a horse race between the exact two-tail probability and the Markov and Mill bounds, we find:



Question: 4.5.7

Let $X_1, \dots, X_n \sim N(0, 1)$. Bound $\mathbb{P}(|\bar{X}_n| > t)$ using Mill's inequality, where $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Compare to the Chebyshev bound.

First note that $\mathbb{E}(\bar{X}_n) = 0$ and $\mathbb{V}(\bar{X}_n) = 1/n$. Thus $\bar{X}_n \sim N(0, 1/n)$. In order for Mill's inequality to be applicable, we transform $\bar{X}_n$ and t to be standard normal:

$$\begin{aligned} \mathbb{P}\left(\left|\frac{\bar{X}_n}{\sqrt{1/n}}\right| > \frac{t}{\sqrt{1/n}}\right) &= \mathbb{P}\left(\left|\frac{\bar{X}_n}{\sqrt{1/n}}\right| > \sqrt{nt}\right) \\ &\leq \sqrt{\frac{2}{\pi}} \frac{e^{-(\sqrt{nt})^2/2}}{\sqrt{nt}} \\ &= \sqrt{\frac{2}{n\pi}} \frac{e^{-nt^2/2}}{t} \end{aligned}$$

And Chebyshev's inequality is given simply by

$$\mathbb{P}(|\bar{X}_n| \geq t) \leq \frac{1}{nt^2}$$

Taking the limit of the ratio of the Mill bound to the Chebyshev for large n gives us

$$\lim_{n \rightarrow \infty} \frac{\sqrt{\frac{2}{n\pi}} \frac{e^{-nt^2/2}}{t}}{1/nt^2} = \lim_{n \rightarrow \infty} \sqrt{\frac{2n}{\pi}} t e^{-nt^2/2} = \boxed{0}$$

yielding the conclusion that Mill is a smaller bound than Chebyshev.